Continuous Electrocardiogram Monitoring with a Loose T-Shirt

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Abstract

Loose-fitting clothing is ideal for unobtrusive electrocardiogram (ECG) monitoring but has long been deemed unviable due to severe motion artifacts caused by fabric shifting. We present a hardware-software co-design that enables reliable cardiac tracking within a standard, loose t-shirt. Physically, the shirt's weight presses a thin sensor against the skin, trapping perspiration to form a stable fluidic anchor via capillary pull and negative pressure, while a structural kirigami pattern mechanically absorbs lateral shear. Computationally, a novel hybrid electrode simultaneously captures ECG and capacitive micro-motion variances to adaptively filter noise during activity. Validated on six participants, the system achieved a 92.7% full waveform detection rate at rest ($r = 0.89$) and an 84.5% R-peak detection rate during walking with a 4.58 bpm heart rate mean absolute error. Users overwhelmingly preferred the shirt over commercial patches and chest straps, proving reliable, long-term health monitoring can be realized in a loose-fitting shirt.

CCS Concepts

• **Human-centered computing** → **Ubiquitous and mobile devices**; • **Applied computing** → **Health care information systems**; • **Hardware** → **Sensor devices and platforms**.

Keywords

Smart Clothing, Wearables, Continuous Electrocardiogram, Motion Artifact Mitigation

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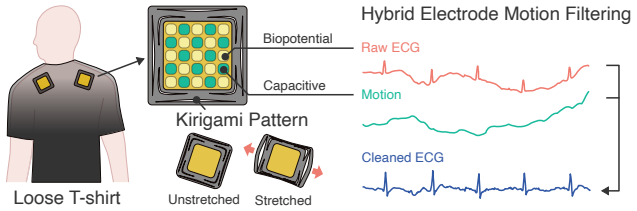


Figure 1: System Overview. Kirigami design for electrode stabilization with a hybrid electrode design for motion filtering.

1 Introduction

Continuous electrocardiogram (ECG) monitoring enables early detection of cardiovascular anomalies missed during standard clinical windows [21]. Moving tracking into everyday life requires embedding dry electrodes into wearables that balance signal quality with user comfort. While adhesive patches and tight athletic straps provide stable contact, they cause skin irritation and are too restrictive for continuous everyday wear [3, 8, 13, 20, 48].

Loose-fitting clothing is the ideal form factor for unobtrusive health tracking. However, common daily movements subject the skin-electrode interface to severe shearing and liftoff, resulting in motion artifacts that distort the signal beyond the capabilities of standard bandpass filters or advanced decomposition techniques. Consequently, loose clothing has long been deemed unviable for reliable biopotential measurement [43].

To resolve the tradeoff between comfort and signal quality, we introduce a hardware-software co-design for continuous ECG monitoring in loose-fitting t-shirts that directly targets the primary sources of noise: perpendicular macro-motions, lateral macro-motions, and micro-motions (Figure 1). Physically, the system utilizes the shirt's natural weight to press a thin, flexible sensor against the upper back, trapping perspiration to establish a gel-free, fluidic suction to anchor it to the skin. Mechanically, a structural kirigami pattern absorbs lateral shear forces to decouple the electrode from fabric shifting. Computationally, an interleaved hybrid electrode simultaneously captures ECG and capacitive micro-motion variances, driving a context-aware software pipeline to adaptively filter mechanical noise during physical activity while preserving waveform geometry at rest.

Evaluating the system on six participants across five activities, we achieved a 92.7% full waveform detection rate with strong morphological correlation ($r = 0.89$) against a research-grade ground

truth at rest. During walking, the system maintained an 84.5% R-peak detection rate and a low heart rate Mean Absolute Error (MAE) of 4.58 bpm. Paired with overwhelming user preference ($\geq 4.89/5.0$) over commercial patches and chest straps, our system proves that reliable, long-term health monitoring can be realized in a standard loose shirt.

Our key contributions are as follows:

- **Motion Decoupling Interface:** An interface that adheres a thin electrode to the skin via the shirt's natural weight and fluidic suction from trapped perspiration, paired with a structural kirigami pattern to decouple the sensor from fabric shear.
- **Hybrid Sensing Pipeline:** A novel hybrid electrode design that simultaneously captures ECG and capacitive micro-motion variances to drive a context-aware software pipeline that adaptively suppresses mechanical noise.
- **Feasibility Evaluation:** A user study with six participants across five activities demonstrating a 92.7% full waveform detection rate at rest ($r = 0.89$), an 84.5% R-peak detection rate during walking, and strong user preference ($\geq 4.89/5.0$) over commercial baselines.

2 Related Work

Prior mobile health research struggles to balance signal quality with user comfort. Traditional ambulatory ECG relies on bulky hardware and irritating adhesives [8, 13, 20]. To bridge this gap, recent works integrate dry electrodes into chest straps [3, 48], clothing [7, 37], furniture [3, 44], car seats [19], and necklaces [46]. However, these systems rely on tight pressure or stationary postures to maintain contact. If clothing is even slightly loose, dynamic fabric shifting induces severe motion artifacts that overwhelm the ECG signal.

Once these artifacts are introduced, existing filtering techniques cannot resolve these extreme disruptions. Bandpass filtering fails when motion noise overlaps the ECG spectrum [2], while decomposition methods like Discrete Wavelet Transform [22] and Empirical Mode Decomposition [23] suffer from mode mixing during movement. Hardware-assisted filtering also presents severe trade-offs: accelerometers [4, 6] capture global body movement rather than localized fabric shifting, while impedance-based tracking [17, 38] requires active current injection. This injection causes signal leakage demanding high sampling rates, and the added circuit complexity elevates power consumption beyond wearable constraints.

Consequently, loose-fitting clothing remains an unviable form factor as neither standard sensing architectures nor filtering algorithms can tolerate the extreme dynamics of fabric shifting.

3 Background and Challenges

ECG measures the heart's electrical activity by detecting differential surface voltages caused by cardiac depolarization and repolarization [12, 29, 40]. The resulting waveform provides critical metrics like heart rate via the RR interval [35] and breathing rate via thoracic baseline variations [31]. Transitioning this monitoring to everyday life requires prioritizing user comfort. While clinical monitors and tight athletic straps are too constrictive for continuous wear, integrating dry electrodes into a loose-fitting t-shirt requires zero behavior change or skin preparation. However, without conductive gel to bridge microscopic air gaps, the dynamic

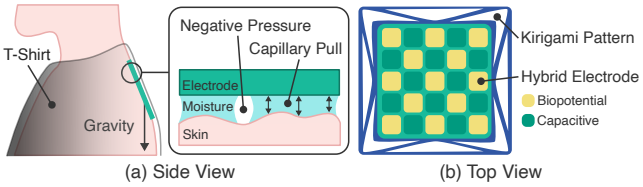


Figure 2: System design overview. (a) Side view: The shirt presses the sensor against the skin, where trapped moisture creates capillary adhesion and a negative pressure seal to secure contact. (b) Top view: An outer kirigami pattern for mechanical motion decoupling surrounds a central hybrid electrode grid for adaptive noise filtering.

movement of loose clothing results in a highly fluctuating skin-electrode impedance [15] due to macro-motion and micro-motion, two distinct mechanical phenomena that introduce layered signal acquisition challenges.

Macro-Motions. These large-scale displacements present the most catastrophic challenge during active daily wear (e.g., walking, reaching). They subject the electrode to severe mechanical forces that heavily disrupt the weakened interface. Lateral shear forces drag the electrode across the skin, introducing massive baseline wander. Simultaneously, perpendicular forces physically lift the electrodes away from the skin surface, widening the air gaps beyond the range of biopotential conduction and completely breaking the electrical circuit. Mitigating macro-motion fundamentally requires localized mechanical intervention to stabilize the fabric and prevent complete contact loss.

Micro-Motions. Even with mechanical stabilization against liftoff, the interface remains susceptible to subtle displacements induced by respiration, postural shifts, or transient fabric twitching. These microscopic movements constantly modulate residual air gaps and contact pressure at the skin, dynamically altering interface impedance and injecting signal distortion. However, from an electromagnetic perspective, these shifting micro-gaps simultaneously alter the localized capacitance between the electrode and the skin. Therefore, while micro-motions disrupt resistive biopotential measurements, they simultaneously produce a deterministic capacitive variance that could theoretically be tracked [9].

4 System Design

Our hardware-software co-design (Figure 2) mitigates motion artifacts by addressing three specific challenges: physically anchoring against perpendicular macro-motions, mechanically absorbing lateral macro-motions, and computationally filtering micro-motions.

Gravity-Assisted Interface for Perpendicular Macro-Motion. Mitigating severe macro-motion requires an anatomical placement that naturally resists perpendicular liftoff. We strategically position thin, non-porous flexible electrodes on the upper back (Figure 2a), providing a stable contact zone compared to the anterior chest, where posture often causes loose clothing to drape away from the skin. This establishes a horizontal bipolar lead mirroring a standard anterior Lead I [25]. We verified this morphological similarity by simultaneously recording from both positions using standard wet Ag/AgCl electrodes. As Figure 3 shows, although the posterior

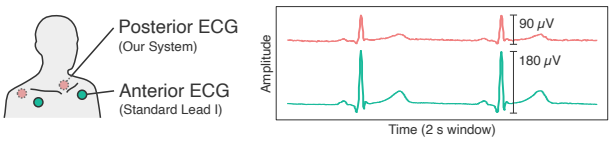


Figure 3: Comparison of raw ECG signals over a 2-second window. The posterior signal (top, red) exhibits a smaller amplitude (90 μV) than the anterior reference (bottom, green, 180 μV) because it is aligned away from the heart axis. Despite this reduction, both signals show aligned R-peaks, and the posterior signal preserves the standard PQRST geometry with a high correlation of 0.98.

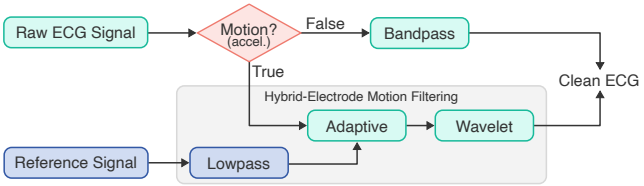


Figure 4: Flowchart depicting the execution flow of the proposed context-aware filtering pipeline. Based on motion evaluation using accelerometer data, a standard bandpass filter is applied during stationary states, whereas a three-stage Hybrid-Electrode Motion Filter, consisting of a lowpass filter, an NLMS adaptive filter, and wavelet decomposition denoising, is utilized when motion is detected.

placement shifts away from the heart's electrical axis which reduce wave amplitudes [47], the waveform strongly resembles a standard Lead I ECG with a high Pearson correlation ($r = 0.98$) with the anterior baseline, preserving essential cardiac morphology. This configuration is highly viable for downstream clinical tasks, such as detecting rhythm-based anomalies like atrial fibrillation (AFib), bradycardia, and tachycardia [33, 41].

Beyond placement, the interface leverages the clothing's weight and fluid dynamics to physically stabilize contact. The shirt's natural drape continuously presses the sensor against the skin, displacing air and trapping natural perspiration. This moisture forms microscopic liquid bridges that draw the substrate and epidermis together via capillary forces [14, 36]. By preventing air reentry, this fluidic layer induces a localized vacuum [5]. Ultimately, gravity prevents perpendicular liftoff, while this fluidic seal anchors against lateral sliding without the need for adhesive.

Kirigami Decoupling for Lateral Macro-Motion. To prevent shifting clothing from dislodging the sensor, the interface uses a kirigami structure to mechanically absorb lateral shear forces (Figure 2b). Originating from the traditional Japanese art of paper cutting, kirigami uses geometric incisions to transform rigid materials into highly stretchable architectures [32, 45]. These strategic cuts allow the substrate's outer regions to freely expand, buckle, and twist out-of-plane when the fabric stretches. By localizing the mechanical strain within the geometric hinges, the structure entirely absorbs external tension before it reaches the core sensing area. Consequently, the central electrode remains undisturbed by the dynamic movement of the loose clothing.



(a) Hybrid Electrode (b) Inside View (c) Back View

Figure 5: Implementation of the proposed system. (a) Hybrid electrode integrated into a stretchable kirigami pattern. (b) Inside view of the prototype showing two sensing electrodes and the central bias electrode. (c) Exterior back view showing the data acquisition system (DAQ).

Hybrid Electrode and Filtering for Micro-Motion. To filter the remaining noise from perpendicular micro-motions that bypass the kirigami structure, the system pairs an interleaved electrode design (Figure 2b) with a context-aware filtering pipeline (Figure 4). Subtle out-of-plane shifts from respiration or posture modulate the skin-electrode distance and induce localized electrochemical artifacts. To physically capture this noise, we split the electrodes into two networks: an exposed active layer to record the biopotential, and a dielectric-isolated reference layer that measures the capacitive variance representing this perpendicular variation. This dual-channel data then feeds into a context-aware adaptive filtering pipeline. To prevent adaptive filters from introducing mathematical artifacts when the reference channel lacks dominant noise [39], the system uses a simple bandpass filter during stationary periods and only triggers the three-stage pipeline during active motion. First, the reference signal undergoes low-pass filtering to extract the low-frequency drift motion and reject any biopotential leakage [10]. A normalized least-mean-squares (NLMS) adaptive filter then uses this reference to dynamically estimate and subtract localized micro-motions from the ECG [42]. Finally, Discrete Wavelet Transform (DWT) decomposition [1] removes any residual high-frequency artifacts to yield the final restored ECG signal.

5 Implementation

We built our prototype using commercially available materials and off-the-shelf hardware (Figure 5). The system comprises three primary modules: (1) the textile sensing interface, (2) the data acquisition hardware, and (3) the signal processing pipeline.

Textile Sensing Interface. The 7.0 × 7.0 cm hybrid electrodes are fabricated from flexible printed circuit boards (FPCBs) with an Electroless Nickel Immersion Gold (ENIG) finish for stable surface conductivity (Figure 5a). These FPCBs are bonded to a spandex substrate and patterned into the 9.5 × 9.5 cm kirigami geometry. Two electrodes are positioned across the upper back to form a horizontal bipolar lead, supplemented by a central fabric bias electrode for common-mode rejection. The complete array is sewn into a standard size Large t-shirt to simulate the loose drape and dynamic shifting typical of everyday wear (Figure 5b).

Data Acquisition Hardware. The electrodes are connected to an OpenBCI acquisition system [34] (Figure 5c). The hardware's analog front-end digitizes both the active biopotential and the capacitive

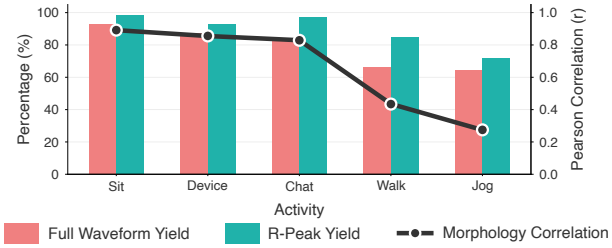
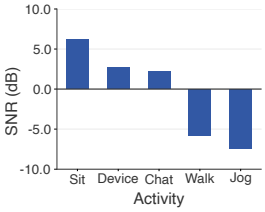


Figure 6: Signal integrity and peak yield across activities. Grouped bars compare the Full Waveform Yield to the R-Peak Yield, while the overlaid line tracks the Pearson correlation (r) of the waveform morphology.

Activity	MAE (ms) P	R	T
Sitting	9.5	5.4	7.7
Using Devices	18.8	8.8	11.6
Chatting	23.5	6.2	15.3
Walking	44.8	12.0	44.5
Jogging	44.8	15.7	47.7

(a)



(b)

Figure 7: System performance metrics across activities. (a) Temporal alignment error for primary waveform peaks. (b) Signal-to-Noise Ratio (SNR) in decibels.

reference channels, wirelessly streaming the raw data to a host computer. The incoming stream is visualized in real-time for signal quality monitoring and logged for offline processing.

Signal Processing Pipeline. Within the software pipeline, the reference channel is first low-pass filtered at 0.5 Hz to isolate mechanical drift. Both channels are then processed through an NLMS adaptive filter (10 taps, $\mu = 0.1$), followed by a DWT utilizing a Daubechies 4 (db4) mother wavelet to extract the cleaned ECG waveform. Finally, we leverage the NeuroKit2 toolbox [26] to compute physiological metrics, specifically, heart rate (HR) and heart rate variability (HRV) derived from the QRS complexes, alongside respiratory rate (RR) via ECG-derived respiration algorithms.

6 Preliminary Feasibility Evaluation

We performed a pilot user study to evaluate our system's performance during simulated daily activities. Six participants without known heart disease wore our custom t-shirt and completed five consecutive 2-minute activities modeling daily behaviors: sitting stationary, using a device while seated, chatting while standing, walking continuously, and jogging in place. The t-shirt continuously recorded the posterior ECG alongside a ground-truth Lead I ECG captured via standard Ag/AgCl electrodes placed at the forearm.

Evaluation Metrics. We evaluated the proposed system across three complementary dimensions:

- **Waveform Morphology:** Assessed against the ground truth via three metrics: (1) *Coverage*, the percentage of matched heartbeats for *R-peak Only* (100 ms window [30]) and *All Peaks* (sequential P, R, T alignment; 120 ms window for P/T waves to accommodate posterior spatial vector shifts [28]). (2) *Structural Similarity*, the Pearson correlation (r) calculated over a 500 ms PQRST window (150 ms pre- and 350 ms post-R-peak) [18]. (3) *Temporal Error*, the mean absolute alignment error (ms) for all matched peaks.
- **Signal-to-Noise Ratio (SNR):** In-band noise variance (in dB), quantified by subtracting an ensemble-averaged heartbeat template from each raw 500 ms window [24].
- **Vital Sign Accuracy:** We report the Mean Absolute Error (MAE) for heart rate (HR) and respiratory rate (RR), alongside standard heart rate variability (HRV) metrics: RMSSD (short-term parasympathetic activity) and SDNN (overall autonomic modulation) evaluated over sliding windows .

6.1 System Performance Across Activities

We apply the evaluation framework to our multi-activity dataset to quantify the system's robustness.

Waveform Morphology Accuracy. As shown in Figure 6, the context-aware pipeline captures clean ECG geometry during stationary states (Sitting, Using Devices, Standing). During Sitting, the system achieves an R-Peak Yield of 98.2%, a Full Waveform Yield (complete PQRST complex) of 92.7%, and a strong morphological correlation ($r = 0.89$) against the ground truth.

However, intense physical activity introduces step impacts, baseline wander, and electromyographic (EMG) noise. During Walking and Jogging, the Full Waveform Yield drops to 65.9% ($r = 0.43$) and 64.5% ($r = 0.27$), respectively, as mechanical noise swamps low-amplitude P and T waves. Despite this peripheral wave corruption, the architecture successfully isolates high-amplitude R-peaks, maintaining an R-Peak Yield of 84.5% during Walking and 71.7% during high-impact Jogging. This motion degradation stems primarily from using a single shirt size. We analyze the relationship between user Body Mass Index (BMI) and performance in Section 6.2 and propose mitigation strategies in the Discussion.

Waveform Timing Precision. Reliable clinical assessment requires temporal precision in peak detection (Figure 7a). During Sitting, our system achieves accurate peak alignment, tracking R-peaks with a Mean Absolute Error (MAE) of 5.38 ms and subtler P and T waves with sub-10 ms precision. In dynamic states, mechanical noise obfuscates lower-amplitude wave boundaries, increasing P and T wave alignment errors to over 44.0 ms. Conversely, R-peak temporal errors remain tightly bounded at 12.0 ms during Walking and 15.7 ms during Jogging. This robust tracking confirms the system maintains accurate beat-to-beat rhythm calculations, even under severe signal deformation from intense physical exertion.

Noise Suppression and SNR. To evaluate signal quality, the post-processed SNR was calculated across all activity protocols (Figure 7b). During stationary activities, the system maintains stable SNR. Sitting achieves an SNR of 6.23 dB, while Using Devices and Standing settle at 2.64 dB and 2.19 dB, respectively, owing to minor postural adjustments and localized muscle contractions. However, dynamic movement introduces massive, low-frequency voltage swings from heel-strikes and torso motion that physically exceed the amplitude of the physiological ECG peaks. These out-of-bound swings completely swamp the true cardiac signal, causing the walking SNR to drop to −5.78 dB. During Jogging, severe sensor

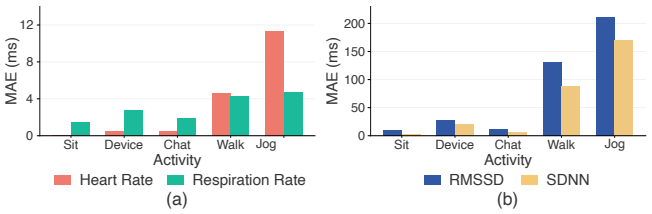


Figure 8: Vital signs accuracy across activities.(a) Cardiorespiratory Rate Errors: Mean absolute error (MAE) for heart rate (HR) and respiration rate (RR). (b) Heart Rate Variability Error: Autonomic tracking errors reported as MAE for RMSSD and SDNN.

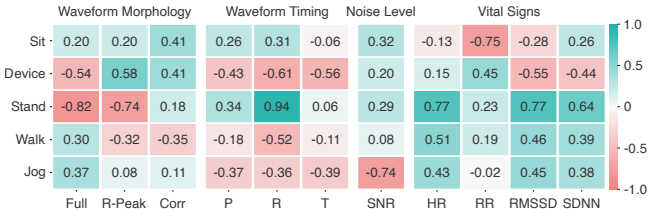


Figure 9: Correlation between user BMI and system performance across activities (red: negative; green: positive). Abbreviations: Full: Full ECG Yield; R-Peak: R-Peak Yield; Corr: Morphology Correlation; P/R/T: peak timing delays; SNR: Signal-to-Noise Ratio; HR/RR: Heart/Respiration Rate; RMSSD/SDNN: HRV parameters.

displacement and EMG interference force the final SNR down to −7.42 dB. This shift into negative SNR values explicitly quantifies the severe degrading impact of physical motion, demonstrating how intense activity introduces a noise floor that physically exceeds the underlying biopotential signal power.

Vital Sign Inference Estimation. We evaluated the system's clinical utility using the vital sign MAE across activities (Figures 8b and 8c). Stationary states yield negligible baseline errors, achieving an HR MAE of 0.03 bpm, RR of 1.44 bpm, RMSSD of 8.98 ms, and SDNN of 3.50 ms during sitting. Minor postural shifts introduce marginal fluctuations, keeping HR errors below 0.51 bpm and HRV metrics within diagnostic boundaries [27]. Conversely, dynamic tasks reveal a clear divergence in error propagation between cardiorespiratory rates and beat-to-beat metrics. Motion artifacts expand HR errors to 4.58 bpm during walking and 11.38 bpm during jogging, remaining near standard industry thresholds ($\leq$ 10 bpm) for consumer tracking [16]. RR demonstrates strong resilience, peaking at 4.25 bpm during walking and 4.75 bpm during jogging, indicating that low-frequency thoracic modulation remains extractable even when high-frequency morphological features are corrupted. Additionally, HRV metrics severely degrade during movement, with RMSSD MAE climbing to 130.90 ms during walking and 211.30 ms during jogging. Because RMSSD relies on the squared differences of successive intervals, minor temporal jitter or missed R-peaks exponentially compounds the error. To help mitigate these errors, we outline potential physical stabilization strategies in our Discussion.

6.2 Impact of User BMI on System Performance

Because our kirigami decoupling mechanism requires physical freedom to deform during movement, its effectiveness depends heavily on individual torso dimensions. Our correlation matrix reveals a distinct, dual-phase relationship between user BMI and system performance (Figure 9). At lower BMIs, the system operates as intended as a looser fit allows the kirigami pattern to successfully decouple outer fabric motion from the skin-anchored electrodes. For users with higher BMIs, the increased torso volume introduces a distinct trade-off, optimizing stationary performance but inducing mechanical failures during motion. During stationary periods, the tighter clothing applies optimal passive contact pressure, securing the electrode and significantly reducing respiration rate errors (Figure 10a: $r = -0.75$).

However, once the user transitions to dynamic movements, this tighter fit causes the decoupling mechanism to fail. Excessive body

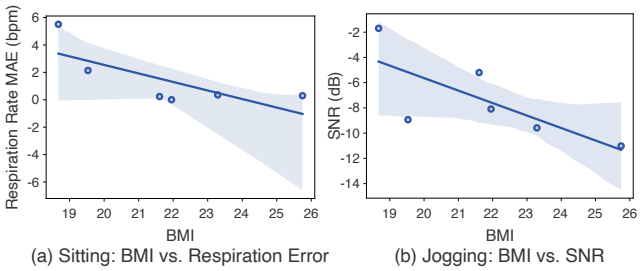


Figure 10: Sensor performance as a function of user BMI during (a) sitting respiration monitoring and (b) jogging. Shaded regions denote the 95% confidence interval ($N = 6$).

volume compresses the structural layers together, generating interlayer friction that pins the kirigami pattern in place. Consequently, when a user stands, gravity and torso extension exert a vertical drag on the tight shirt, pulling the coupled electrode into a suboptimal position. This reduces Full ECG Yield ($r = -0.82$) and drives up R-wave timing errors ($r = 0.94$). While mild transitions like walking can temporarily break this friction lock and allow the sensor to self-adjust, high-intensity movements completely overwhelm the interface. During jogging, violent dynamic shear forces bypass the kirigami structure entirely and map directly onto the coupled electrode, causing severe SNR degradation (Figure 10b: $r = -0.74$).

6.3 Ablation Study

Empirical results highlight a critical trade-off in physiological signal modeling: aggressive motion artifact mitigation is essential during dynamic states but systematically distorts signals during rest. As shown in Table 1, applying the standalone motion filtering pipeline during the Sitting scenario degrades morphology, dropping signal correlation from 0.890 to 0.751 and inflating heart rate MAE tenfold compared to a simple bandpass baseline. Conversely, relying solely on static baseline filtering fails during intense activities like Jogging, yielding an unacceptable MAE of 32.00 bpm. Our context-aware framework resolves this dilemma by using real-time inertial data to dynamically activate aggressive filtering only when physical artifacts threaten signal integrity. This approach successfully preserves undisturbed signals during rest while robustly mitigating noise during motion (complete metrics are documented in Appendix A).

Table 1: Ablation Analysis of Context-Aware Balancing under Extreme Scenarios

Scenario	Configuration	Coverage (%)	Morph. Corr (r)	HR MAE (bpm)
Stationary (Sitting)	Bandpass Baseline	92.6	0.890	0.03
	Ours (Motion Filter)	91.7	0.751	0.45
	Ours (Context-Aware)	**92.7**	**0.890**	**0.03**
High-Motion (Jogging)	Bandpass Baseline	50.6	0.256	32.00
	Ours (Motion Filter)	64.0	0.272	11.59
	Ours (Context-Aware)	**64.5**	**0.272**	**11.38**

6.4 Usability

Upon completion, participants evaluated the proposed system via a 5-point Likert scale and open-ended feedback. Quantitative results demonstrated overwhelming acceptance: metrics for comfort, convenience, and ease of use (Q1–Q3) received exclusively positive responses (>= 4.89/5.0). Furthermore, participants unanimously preferred our system over commercial ECG patches (Q4) and chest straps (Q5), rating the overall experience (Q6) exceptionally high. Moreover, qualitative feedback echoed these quantitative findings. Users praised the zero-effort, frictionless sensing of an everyday t-shirt that successfully bypasses the skin irritation and physical restriction of traditional monitors. The sole critique noted the data-acquisition module's bulkiness, a footprint easily resolved in future iterations via custom PCB miniaturization.

7 Discussion and Future Work

While our system demonstrates the feasibility of ECG capture using loose-fitting clothing, transitioning this prototype into an everyday wearable requires addressing several remaining limitations. Future development must optimize the entire system from the clothing design to the companion software, while expanding validation across diverse demographics.

Hardware Optimization. To directly address the motion-induced degradation observed during dynamic tasks, future iterations will focus on refining both clothing form factor and sensor geometry. First, because our empirical results (Sec. 6.2) reveal that user BMI dictates system motion resilience, a straightforward mitigation strategy is to fabricate multiple clothing sizes. However, because standardized sizing is discretized, it cannot perfectly accommodate every individual. We must therefore explore novel structural designs, such as low-friction interlayer linings or localized elastic zoning across the torso, that allow a single size to tolerate a wider range of BMIs without compromising the kirigami mechanism during movement. Second, we will optimize the electrode configuration by exploring multi-electrode clusters and alternative geometries to maintain localized skin contact during movement. This multi-sensor approach enables an adaptive channel-selection strategy that dynamically evaluates each element's SNR, automatically routing the most stable channel to bypass motion-induced noise. Third, we will strategically alter the surface area ratios within our hybrid structure to leverage the distinct physical properties of each sensor type. Because capacitive components are highly sensitive, we can reduce their individual surface area and allocate that physical space to the biopotential elements. Expanding this direct contact area lowers the characteristically high contact impedance of dry electrodes [11], yielding a better signal quality. Finally, miniaturizing the processing unit and routing its connections to the electrodes via an integrated FPCB will replace loose wires to mitigate motion-induced cable noise, while simultaneously improving system portability and user comfort during long-term deployment.

Real-time Processing and Monitoring. We plan to develop a mobile companion application for real-time ECG visualization and automated signal quality assessment. This application will handle continuous signal streaming to a smart device, dynamically flagging electrode displacements or motion artifacts to trigger adaptive edge processing. Building upon this real-time visualization, we will extend the pipeline beyond basic heart rate to estimate additional vital signs, including respiratory rate and heart rate variability (HRV), to provide immediate health feedback. Furthermore, accumulating this continuous stream over time will allow us to deploy advanced analytical algorithms to monitor long-term cardiovascular trends and automatically detect early signs of heart-related diseases in everyday scenarios.

Extensive User Study. Following the hardware and software optimizations, we plan to evaluate the system through a more extensive user study encompassing diverse demographics. Because clothing fit and skin contact are profoundly influenced by body morphology, future validation will focus specifically on a wide distribution of user BMIs to thoroughly test our sizing models against distinct body shapes. Furthermore, we will introduce longitudinal studies where participants wear the clothing continuously for a full day. This long-term monitoring will allow us to investigate if and when signal degradation occurs over time, verifying system resilience across both stationary intervals and high-intensity activities. Additionally, to ensure clinical viability, we will benchmark our system against FDA-cleared clinical-grade monitors, such as a standard ambulatory Holter system, moving beyond our current forearm-referenced wet electrode baseline. These real-world evaluations will help define the boundaries of the prototype's robustness and guide future design iterations.

8 Conclusion

We introduce a hardware-software co-design for continuous, unobtrusive ECG monitoring within standard, loose-fitting t-shirts. By embedding kirigami-stabilized hybrid electrodes on the upper back, the system mechanically decouples the interface from everyday clothing disruptions. Paired with a context-aware filtering pipeline, this architecture balances stationary precision with dynamic noise suppression. Evaluated across five activities, the system achieved a 92.7% waveform detection rate ($r = 0.89$) at rest, and an 84.5% R-peak detection rate with a 4.58 bpm heart rate MAE during walking. Strongly preferred by users ($\geq 4.89/5.0$), this zero-effort system proves that reliable, long-term cardiovascular monitoring can be comfortably deployed in the wild.

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Table 2: Comprehensive Algorithmic Benchmarking and Ablation Matrix Across All Scenarios

		Coverage (%)		Morph.	SNR	Timing MAE (ms)			Vital Sign MAE			
Scenario	**Pipeline Configuration**	**Full**	**R-Peak**	**Corr (*r*)**	**(dB)**	**P**	**R**	**T**	**HR (bpm)**	**RR (bpm)**	**RMSSD (ms)**	**SDNN (ms)**
Sitting	Bandpass	92.6%	**98.2%**	**0.890**	6.20	9.5	5.4	**7.7**	0.03	1.44	**9.0**	**3.5**
	Wavelet	91.1%	**98.2%**	0.877	5.53	10.5	5.3	7.9	**0.03**	1.41	9.6	3.8
	Adaptive	88.2%	97.6%	0.729	1.84	12.1	5.1	20.2	0.64	0.98	18.4	13.5
	Ours (*Motion Filter Only*)	91.7%	97.8%	0.751	**7.41**	12.0	**5.1**	20.9	0.45	**0.90**	15.4	10.9
	Ours (Context-Aware)	**92.7%**	**98.2%**	**0.890**	6.23	**9.5**	5.4	**7.7**	0.03	1.44	**9.0**	**3.5**
Using Devices	Bandpass	**87.3%**	93.0%	0.852	2.56	**18.8**	8.8	11.9	0.49	2.78	27.2	20.2
	Wavelet	86.7%	**93.2%**	0.836	2.15	19.9	8.8	12.2	**0.26**	2.17	**21.9**	**14.7**
	Adaptive	84.2%	90.9%	0.705	1.74	19.3	8.0	25.7	1.37	**2.04**	26.1	25.3
	Ours (*Motion Filter Only*)	86.8%	93.1%	0.713	2.45	19.5	**7.9**	26.2	0.88	2.06	20.0	17.7
	Ours (Context-Aware)	87.1%	93.0%	**0.853**	**2.64**	**18.8**	8.8	**11.6**	0.49	2.77	27.2	20.2
Standing	Bandpass	**84.5%**	96.9%	0.830	**2.27**	**23.4**	6.2	15.3	**0.51**	1.90	11.9	5.8
	Wavelet	**84.5%**	**97.0%**	0.819	2.08	24.8	**6.2**	15.9	**0.51**	**1.73**	**11.6**	**5.7**
	Adaptive	71.0%	94.6%	0.662	-0.36	29.7	6.9	29.4	1.72	2.07	29.3	25.9
	Ours (*Motion Filter Only*)	77.8%	96.1%	0.707	1.21	28.0	7.1	29.4	0.85	2.72	22.1	14.6
	Ours (Context-Aware)	84.3%	**97.0%**	**0.831**	2.19	**23.4**	6.2	**15.3**	**0.51**	1.85	11.9	5.8
Walking	Bandpass	56.3%	73.0%	0.359	-10.19	**40.0**	**9.3**	**36.3**	15.91	**3.45**	752.1	506.2
	Wavelet	55.1%	72.4%	0.347	-10.68	41.9	9.4	34.9	17.12	3.23	979.9	694.3
	Adaptive	60.8%	82.8%	0.382	-8.89	44.7	11.8	42.2	6.44	3.81	228.5	155.5
	Ours (*Motion Filter Only*)	65.7%	83.9%	**0.437**	-6.61	43.6	12.2	41.6	5.07	4.05	154.7	101.6
	Ours (Context-Aware)	**65.9%**	**84.5%**	0.433	**-5.78**	44.8	12.0	44.5	**4.58**	4.25	**130.9**	**87.6**
Jogging	Bandpass	50.6%	51.6%	0.256	-9.74	46.0	18.0	**39.3**	32.00	5.02	609.3	471.0
	Wavelet	53.8%	52.1%	0.245	-10.26	45.4	18.4	41.4	31.75	**4.56**	607.5	466.4
	Adaptive	**65.1%**	70.7%	0.246	-8.94	**43.7**	15.5	48.0	12.10	4.85	234.4	184.0
	Ours (*Motion Filter Only*)	64.0%	71.5%	**0.272**	-9.18	45.4	**15.2**	47.6	11.59	5.23	**211.1**	**169.1**
	Ours (Context-Aware)	64.5%	**71.7%**	**0.272**	**-7.42**	44.8	15.7	47.7	**11.38**	4.75	211.3	170.4

A Full Ablation Study

Table 2 provides a comprehensive performance breakdown of different configurations across the evaluated activity scenarios. We benchmark our context-aware architecture against standard baselines including conventional bandpass filtering, wavelet denoising, and adaptive noise cancellation. We also include an ablated variant of our pipeline utilizing motion filtering without context awareness. Performance is evaluated across signal availability (Coverage), structural preservation (Morphology Correlation), window tracking accuracy (Timing MAE), and downstream vital sign inference (HR, RR, RMSSD, and SDNN MAE). For each activity block, the top-performing configuration within each metric is highlighted in bold.

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